

■ RISING WATERS

A Machine Learning Approach to Flood Prediction

A guided internship project report submitted for evaluation

Project Domain:	Artificial Intelligence & Machine Learning
Program:	SmartInternz Long-Term Internship Program
Evaluation Platform:	SkillWallet Dashboard
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Date:	July 2026

State Council of Higher Education (APSCHE)
In collaboration with SmartBridge & IBM Cloud

1. Abstract & Introduction

Floods are among the most destructive natural disasters, causing severe losses to lives, agriculture, and infrastructure annually. This project, titled Rising Waters, addresses the challenge of early flood warnings by building an intelligent machine learning-powered classification pipeline. Utilizing meteorological inputs including rainfall patterns, humidity, temperature, visibility, and atmospheric pressure, multiple models are trained, compared, and deployed. The final best-performing classifier (XGBoost) achieves an accuracy score of 96.55% and is integrated into a dynamic, responsive Flask web application to support real-time evacuation and disaster preparedness decisions.

Climate change has triggered irregular weather cycles, resulting in erratic, high-volume seasonal rainfall and severe flash floods. Conventional early warning models fail to parse non-linear environmental variables in a timely manner. To solve this, predictive machine learning models can process complex weather data instantly. This report documents the end-to-end implementation of the Rising Waters system, detailing dataset generation, robust preprocessing pipelines, multi-model evaluation, and the final web deployment.

2. Data Pre-Processing Pipeline

Raw environmental data is frequently incomplete or subject to sensor noise. The pre-processing pipeline in this project guarantees model stability and high accuracy through the following sequential operations:

2.1 Handling Missing Values:

Missing data in numerical features (such as temperature and visibility) are imputed using the column median to prevent feature bias. Any missing values in categorical fields are imputed using the mode.

2.2 Outlier Detection and Treatment:

Extreme records caused by sensor malfunctions are detected using the Interquartile Range (IQR) method. Values falling beyond 1.5 IQR limits are capped at the lower and upper bounds. This preserves the dataset size while removing gradient distortions.

2.3 Categorical Encoding:

The meteorological category Weather_Condition (consisting of values: Sunny, Cloudy, Overcast, Rainy) is encoded into binary variables using One-Hot Encoding. This expands the input dimensions from 7 numerical features to 11 independent inputs, enabling the models to learn sky-condition coefficients.

2.4 Feature Scaling and Splitting:

To standardize gradient descents, features are scaled to zero-mean and unit-variance using the StandardScaler. The processed dataset (consisting of 1450 records) is divided into an 80% training set (1160 samples) and a 20% test set (290 samples), stratified by the target label to maintain uniform class distributions.

3. Model Building & Evaluation

Four classification models are trained on the stratified training set and evaluated on the test set. A summary of the model evaluation results is presented below:

Model	Test Accuracy	Classification Characteristics
Decision Tree	95.86%	Learns hierarchical splits; fast but prone to overfitting.
Random Forest	96.90%	Bagging ensemble; highly resilient to variance.
K-Nearest Neighbors	93.79%	Distance-based; slow on large dimensions.
XGBoost (Selected)	96.55%	Gradient boosted trees; optimal trade-off and calibration.

XGBoost Confusion Matrix:

On the 290 test samples, the production-selected XGBoost model demonstrates high precision and recall, making only 1 false-positive mistake:

- True Negatives (No Flood): 196 samples correctly predicted.
- False Positives (Type I Error): 1 sample incorrectly predicted as flood.
- False Negatives (Type II Error): 9 samples missed.
- True Positives (Flood Warning): 84 samples correctly predicted.

4. Flask Application Architecture

The finalized XGBoost classifier and the fitted StandardScaler are serialized to floods.save. We built an interactive, responsive Flask-based web application to deploy this model for production, providing real-time predictions and flood threat metrics to users.

5. Conclusion & Recommendations

The Rising Waters system successfully combines advanced tree-boosting architectures (XGBoost) with user-friendly web dashboards (Flask) to deliver accurate, localized flood probability outputs. Achieving a test accuracy of 96.55%, the model is highly reliable for operational planning. We recommend integrating this API with telemetry sensor grids (measuring rainfall and pressure rates) to automate flood warnings directly to the public.